Appendix 7. Sports activity categorized into levels of sports participation, time spent playing sports (including session length and frequency), and impact and intensity.

	Characteristics*	Time after surgery	Participation rate	Time spent playing sports	Frequency	Session length	Intensity level
Total hip arthroplasty							
General sports							
Bonnin et al. 2018	N: 1042 Age: 60.6 ± 8.8 57% male BMI: 26.6 ± 4.6	44.4 ± 14.0 months	91% participated regularly or frequently in sports				2.6% competitive
Bonnin et al. 2020	N: 259, posterolateral approach Age: 61 ± 7.8 (39-78) 63% male BMI: 27 ± 4.2 (17-41)	44 ± 14.1 (24-74) months	81% participated regularly or frequently in sports				
	N: 259, anterolateral approach Age: 62±8.1 (21-76) 59% male BMI: 27±4.6 (18-45)	43±14.5 (24-79) months	81% participated regularly or frequently in sports				
Breuer et al. 2020	N: 55 Age 61±10 53% male BMI: 27.9±4.9	38±4.6 months	83%†				
Donner et al. 2019	N: 51 Age: 63.1 (36.7-76.8) 56.9% male BMI (median): 27.6 (16.6-41.8) 100% bilateral	4.9 (4.3-5.8) years	76.5% participated regularly in sports	4.2 hours/wk	3.4±2.9 sessions/wk		
Hara et al. 2018	N: 524 Age: 62.9±10.1 16% male BMI: 22.9±3.3 19% bilateral	68.2 months	30.5% participated at least once a month in sports				

	Characteristics*	Time after surgery	Participation rate	Time spent playing sports	Frequency	Session length	Intensity level
Heiberg et al. 2016	N: 30 Age: 70.5±8.4 56.7% male	5 years			3±1.0 sessions/wk		
Innmann et al. 2016	N: 86 Age: 52 (21-60) 61% male BMI: 27 (18-39) 4% bilateral	11 (10-12) years	79%†		2.6 sessions/wk	55 min	
Jassim et al. 2019	N: 40 Age: 53.1±8.4 47% male	3.3±1.1 years			7.5% 1 session/wk 43% 2-3 sessions/wk 45% 4-6 sessions/wk 5% daily		
Madrid et al. 2019	N: 535 Age: 67 (13 -91) 31% male BMI: 25.5±3.9	>1 year	7.1%† participated in institutionalized sports	42% <5 hrs/wk 37% 5-10 hrs/wk 21% 10-20 hrs/wk 0% >20 hrs/wk			
Ollivier et al. 2014	N: 571 Age: 61.3±10.9 Gender: 52% male BMI: 27±3.2 0% bilateral	9.8±2.9 years	64%†				
Ortmaier et al. 2017	N: 137 Age: 65.6±12.4 BMI: 26.6±4	20.4±2.3 (18-22) months	87%†		18% 0 sessions/wk 12% 1 session/wk 20% 3 sessions/wk 9% 4 sessions/wk 27% >4 sessions/wk	24% 0-15min 9% 15-30 min 15% 30-60 min 22% 60-120 min 29% >120 min	>80% recreational
Payo-Ollero et al. 2020	N: 46 Age: 41 (37-48) 72% male BMI: 26.1 (24.5-29) 25% bilateral	7.5 (1-11) years	78%†				
Schmidutz et al. 2012	N: 68 Age: 55±12 Gender: 60% male BMI: 26±4 kg/m2 12% bilateral Short-stem hip arthroplasty	2.7±0.7 (2.0-4.2) years	91%†		1.8±1.1 sessions/wk	66±33 min	

	Characteristics*	Time after surgery	Participation rate	Time spent playing sports	Frequency	Session length	Intensity level
Smith et al. 2018	N: 105 Age: 68.2±9.3 42.9% male BMI: 28.8±4.2	12 & 24 months		22.2% <1 hr/wk 0% 1-2 hrs/wk 66.7% 2-4 hrs/wk 11.1% >4 hrs/wk 11.1% <1 hr/wk 11.1% 1-2 hrs/wk 33.3% 2-4 hrs/wk 45.4% >4 hrs/wk 36.8% <1 hr/wk 47.4% 1-2 hrs/wk 15.8% 2-4 hrs/wk 0% >4 hrs/wk	87.5% never 9.7% 1/2 sessions/wk 2.8% 3/4 sessions/wk 0% 5-7 sessions/wk 94.4% never 2.8% 1/2 sessions/wk 1.4% 3/4 sessions/wk 1.4% 5-7 sessions/wk 73.6% never 4.2% 1/2 sessions/wk 13.9% 3/4 sessions/wk 8.3% 5-7 sessions/wk		Light intensity Moderate intensity Strenuous intensity
Judo							
Lefevre et al. 2013	N: 27 judokas Age: 63±7.2 33% bilateral	8.8±7.1 years	81.5%†		2.5±1 sessions/wk		0% competitive
Hip resurfacing							
General sports							
Amstutz et al. 2019	N: 661 Age: 51.9 (14-78) 70% male BMI: 26.5 (16.7-46.5)	10.1 (1-16) years		Activity quantity (frequency†duration): Frequency: 1: 1-4x/month; 2: 5-8x/month; 3: 9-12x/month; 4: >12x/month; Duration 1: 0-30min; 2: 30-60min; 3: 60-120min; 4: >120min 12.4			
Banerjee et al. 2010	N: 138 Age: 52.6 (38-71) Male hips: 59% 10% bilateral	23.5 (12-42) months	97%†, 61% daily				
Fisher et al. 2011	N: 117 Age: 54 (30-73) 57% male 14% bilateral	30 (16-50) months	73% participated in sports at least once a month				

	Characteristics*	Time after surgery	Participation rate	Time spent playing sports	Frequency	Session length	Intensity level
Girard et al. 2013	N: 50 participating in high-impact sport preop Age: 51.5 (30.8-64.8) 90% male BMI: 23.7 (21.7-33.6) 10% bilateral	44.1 (39.1-54.5) months	98%†	2.9 (2-9) hrs/wk			
Le Duff & Armstutz, 2011	N: 201 Age: 49.6 74.6% male BMI: 26.9 (19-46) 28% bilateral	1.8 (1.0-4.9) years, 9.1 (5.0-13.4) years	1.8yr: 96% 9.1yr: 90% in at least one sporting activity		1.8yr: 33.9% >12 sessions/month 9.1yr: 33.1% >12 sessions/month	1.8yr: 45% sessions 30-60 min; 9.1yr: 36% sessions 30-60 min	1.8yr: 5.5% competitive 9.1yr: 10.1% competitive
Sandiford et al. 2015	N: 79 active patients Age: 54.9 (3405-73.6) 67% male BMI: 25.2 (19.8-31.9) 1% bilateral	8.5 (8-10) years	97%†		100% >3 sessions/wk	100% 60-90 min	14% competitive
Jogging							
Fouilleron et al. 2012	N: 40 runners preop Age: 50.7 (31-61) 90% male BMI: 24.8 (21.7-33.6) 8% bilateral	33.3 (26-41) months	91.6%†	3.1 hrs/wk			s: 28.4 km/wk
Total knee arthroplasty							
General sports							
Hepperger et al. 2018	N: 200 Age: 72±7.7 40% male BMI: 27.2±5.0 17.5% bilateral	1 year 2 years	82% at least occasionally, 69% at least2x/week 83% at least occasionally, 70% at least 2x/week		18% no sports 13% occasionally 59% 2-3 sessions/wk 10% >5 sessions/wk 17% no sports 13% occasionally 57% 2-3 sessions/wk 13% >5 sessions/wk		
Ho et al. 2016	N: 39 Age: 59 (57-64) 30% male BMI: 32.5 [28.8-38.4] 3% bilateral	4.0±1.2 years	70%†				

	Characteristics*	Time after surgery	Participation rate	Time spent playing sports	Frequency	Session length	Intensity level
Jassim et al. 2019	N: 24 Age: 60±2.5 42% male	3±0.9 years			17% once a week 25% 2-3 sessions/wk 46% 4-6 sessions/wk 13% daily		
Mayr et al. 2015	N: 81 Age at FU: 71.8±5.4 47% male BMI: 28.4	6.4±0.9 years		5.3 hrs/wk	3.5 sessions/wk		
Plassard et al. 2020	N: 443 Age: 69 (41-90) 36.6% male BMI: 29.3 (19-46)	43 [23-49] months	85%†				
Ristolainen et al. 2019	N: 70, knee injury as cause of TKA Age: 51.8±5.2 BMI: 29.0±5.0	6.2±3.7	All: 78% Sports injury group: 89% Non-sports injury group: 74% Regular exercise		>50% exercised >10x/month	>1 hour for 50% of patients	Total: 68% walking intensity, 14% high-intensity activity
Smith et al. 2018	N: 116 Age: 67.3±8.3 41.4% male BMI: 30.1±4.9	12 & 24 months		0% <1 hr/wk 12.5% 1-2 hrs/wk 56.3% 2-4 hrs/wk 31.2% >4 hrs/wk 38.5% <1 hr/wk 15.4% 1-2 hrs/wk 46.1% 2-4 hrs/wk 0% >4 hrs/wk 51.4% <1 hr/wk 34.3% 1-2 hrs/wk 14.3% 2-4 hrs/wk 0% >4 hrs/wk	84.5% never 13.6% 1/2 sessions/wk 4.9% 3/4 sessions/wk 0% 5-7 sessions/wk 87.4% never 6.8% 1/2 sessions/wk 4.9% 3/4 sessions/wk 0.9% 5-7 sessions/wk 66.0% never 8.7% 1/2 sessions/wk 18.4% 3/4 sessions/wk 6.9% 5-7 sessions/wk		Light Moderate Strenuous
Vielgut et al. 2016	N: 236 Age: 62.7±11.4 18% male 10% bilateral	14.9±3.0 years	70.8% Participated regularly in sports				
Jogging							
Ho et al. 2016	N: 39 Age: 59 (57-64) 30% male BMI: 32.5 [28.8-38.4] 3% bilateral	4.0±1.2 years	2% (n=1)		15 sessions/wk	60 min	

	Characteristics*	Time after surgery	Participation rate	Time spent playing sports	Frequency	Session length	Intensity level
Tennis							
Ho et al. 2016	N: 39 Age: 59 (57-64) 30% male BMI: 32.5 [28.8-38.4] 3% bilateral	4.0±1.2 years	2% (n=1)				
Other							
Ho et al. 2016	N: 39 Age: 59 (57-64) 30% male BMI: 32.5 [28.8-38.4] 3% bilateral	4.0±1.2 years	Golf: 18% Bowling: 0% Cycling: 38% Swimming: 22% Hiking: 12% Dancing: 28% Basketball: 0% Other: 18%		1 [1-3] sessions/wk 12 [4-15] sessions/wk 4 [3.5-13] sessions/wk 4 [1.2-12] sessions/wk 1.5 [0.9-4.5] sessions/wk 12 [8-30]	60 [60-120] min 30 [20-40] min 30 [30-45] min 30 [20-60] min 45 [15-60] min 90 [55-180] min	
Lefevre et al. 2013	N: 8 judokas Age: 72.8±5.2 25% bilateral	7±5.9 years	62.5%†		2.5±1 sessions/wk		0% competitive
Pioger et al. 2020	N: 143 active golfers Age: 65.7±(43-85) 85.3% males Bilateral: 0%	4.6 (2.0-14.2) years		10.2±6.6 hrs/wk			Handicap: 24.4±13.3
Unicompartimental knee arthroplasty							
General sports							
Canetti et al. 2018	N: 11, UCLA ≥5 preop Age: 66.5±6.8 BMI: 24.2±4.3 18% male Surgery: Robotic UKA	34.4±10.5 months	100%†				
	N: 17, UCLA ≥5 preop Age: 59.5±9.9 BMI: 26.3±3.8 28% male Surgery: normal UKA	39.3±15.5 months	94%†				
Ho et al. 2016	N: 33 Age: 60 (53-64) 33% male BMI: 30.3 [27.6-33.7] 9% bilateral	4.0±1.2 years	72%†				

	Characteristics*	Time after surgery	Participation rate	Time spent playing sports	Frequency	Session length	Intensity level
Jahnke et al. 2015	N: 135 Age: 67.1 (38-88) 53.4% male 9% bilateral	2±1.47 years	93.2%, 87% weekly				
Kim et al. 2019	N: 42 Age: 63.6±5.5 17% male BMI: 25.3±2.4 0% bilateral	24 months	76.2%†		3.0±1.3 sessions/wk	1.3±0.7 hours/session	
Kleeblad et al. 2020	N: 164 Age: 62.3±8.8 55% male BMI: 27.6±4.4 9% bilateral	20.2 months	89.6%†				
Lo Presti et al. 2019	N: 53, did ≥1 sport preop Age: 59.7 (46-66) 28.3% male	48±6 (24-72) months	90%†		1.9 sessions/wk	43 min	100% recreational
Panzram et al. 2018	N: 27 Age: 62.5±8.3 (49-76) 56% male 11% bilateral	60±8.3 (47-69) months	96%†				
Panzram et al. 2020	N: 177 Age at FU: 64.4±9.7 (38-82) BMI at FU: 30.9±5.4 8% bilateral	37.1±9.8 (24-60) months	83.4%†		2.8±1.4 sessions/wk 15.6% >4 sessions/wk ≥60 min (%)		
Pietschmann et al. 2013	N: 131 Age: 65.3 (44-83) 44% male Bilateral: 6%	4.2 (1-10) years	60%†				
Walker et al. 2015a (return to sports)	N: 45 Age: 60.1±10.5 42% male BMI: 27 0% bilateral	3 (2.0-4.3) years	95.6%†		3 sessions/wk	44% >1 hour/session	
Walker et al. 2015b (Sports, PA)	N: 93 Age: 55 (36-60) 48% male BMI: 32 (20-58) 17% bilateral	4.4±1.6 (2.3-8.4) years	91%†		53% ≥3 sessions/wk	45% ≥1 hour/sessions	

	Characteristics*	Time after surgery	Participation rate	Time spent playing sports	Frequency	Session length	Intensity level
Zimmerer et al. 2021	N: 19 Age: 26.7±10.5 (36.0-76.0) 26% male BMI: 27.2±5.7 (17.3-38.0)	4.6±2.2 (2.0-9.7) years	68%†		2.3 sessions/wk	45.9 min/session	
Jogging							
Ho et al. 2016	N: 33 Age: 60 (53-64) 33% male BMI: 30.3 [27.6-33.7] 9% bilateral	4.0±1.2 years	6% (n=2)		10 (8-12)	32 (20-45)	
Jahnke et al. 2015	N: 135 Age: 67.1 (38-88) 53.4% male 9% bilateral	2±1.47 years	24%		2.22 (0: 0 sessions/month 1: 1 sessions/month 2: 2 sessions/month 3: 1 session/wk 4: 2-3 sessions/wk 5: 2-3 sessions/wk 6: >4 sessions/wk)	2.16 (0= 0 min 1= <15 min 2= 15-60 min 3= 61-120 min 4= 121-180 min 5= >3 hrs)	
Tennis							
Ho et al. 2016	N: 33 Age: 60 (53-64) 33% male BMI: 30.3 [27.6-33.7] 9% bilateral	4.0±1.2 years	0%				
Jahnke et al. 2015	N: 135 Age: 67.1 (38-88) 53.4% male 9% bilateral	2±1.47 years	6%		1.92 (0: 0 sessions/month 1: 1 session/month 2: 2 sessions/month 3: 1 session/wk 4: 2-3 sessions/wk 5: 2-3 sessions/wk 6: >4 sessions/wk)	1.62 (0 = 0 min 1= <15 min 2= 15-60 min 3= 61-120 min 4= 121-180 min 5= >3 hrs)	

	Characteristics*	Time after surgery	Participation rate	Time spent playing sports	Frequency	Session length	Intensity level
Other							
Ho et al. 2016	N: 33 Age: 60 (53-64) 33% male BMI: 30.3 [27.6-33.7] 9% bilateral	4.0±1.2 years	Golf: 31% Bowling: 8% Cycling: 36% Swimming: 14% Hiking: 42% Dancing: 6% Basketball: 3% (n=1) Other: 8%		4 [0.2-20] sessions/wk 3 (2-4) sessions/wk 8 [4-21] sessions/wk 4 [2.2-28.5] sessions/wk 4 [1-16] sessions/wk 6 [4-8] sessions/wk 1 session/wk 16 [12-16] sessions/wk	30 [30-60] min 100 (60-180) min 30[20-40] min 45 [17.5-60] min 60 [45-60] min 60 [60-60] min 60 min 45 [20-60] min	
Jahnke et al. 2015	N: 135 Age: 67.1 (38-88) 53.4% male 9% bilateral	2±1.47 years			Hiking: 3.02 Swimming: 2.37 Cycling: 3.34 CC skiing: 1.40 DH skiing: 0.58 Golf: 2.89 Dancing: 0.87 Tennis: 1.92 Ball games: 0.55 Other: 3.47 (0:0 sessions/month 1: 1 session/month 2: 2 sessions/month 3: 1 session/wk 4: 2-3 sessions/wk 5: 2-3 sessions/wk 6: >4 sessions/wk)	Hiking: 3.18 Swimming: 2.96 Cycling: 2.73 CC skiing: 2.20 DH skiing: 0.74 Golf: 4.00 Dancing: 1.13 Tannis: 1.62 Ball games: 0.55 Other: 2.88 (0 = 0 min 1= <15 min 2= 15-60 min 3= 61-120 min 4= 121-180 min 5= >3 hrs)	
Mixed arthroplasty							
General sports							
Naylor et al. 2019	N: 1289 Age: 67.2±9 45% men BMI: 30.9±6 5% bilateral	3 years	THA: 67.3% ≥1 session/week TKA: 60.6% ≥1 session/week				
Pisanu et al. 2020	N: 118 Age: 73 [67-77] 28.7% male BMI: 29.1±4.7 3% bilateral	3.1±1.13 years	87%†				

	Characteristics*	Time after surgery	Participation rate	Time spent playing sports	Frequency	Session length	Intensity level
Jogging							
Abe et al. (2014)	Age: 63.8±11.7 55% male BMI: 29.6±5.6	4.8 (2.3-7.8) years	3.8% participated in jogging at least 1x/week		4 (1-7) sessions/wk	29 (5-90) min	v: 7.7 (3-18) km/hour s: 3.6 (0.5-15) km

*At time of surgery, unless indicated otherwise; † Definition not specified

Abbreviations: BMI: body mass index; FU: follow-up; HRA: hip resurfacing arthroplasty; THA: total hip arthroplasty; TKA: total knee arthroplasty; UKA: unicompartimental knee arthroplasty